Atlantium®
Hydro-Optic Water Disinfection System

User Interface Operator's Guide

RZ163 Series

Atlantium
Illuminating Water Technologies

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- Publication Catalog No. PBF003500, Revision A
- System and accessory specifications subject to change without notice

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INTRODUCTION

This guide offers detailed instructions for the routine operation of the Atlantium RZ163 hydro-optic water disinfection systems, incorporating three or more UV lamps in one or two reactor branches.

The system is designed to be operated from the color touch-screen LCD monitor panel located in the front door of the system's electrical cabinet (see Figure 2).

Important Note

- Operating instructions for the RZ163-11 (single-lamp) and RZ163-12 (dual-lamp) systems can be found in the User Interface Operator's Guide (P/N # PB0002500).
- Not all of the features described in this guide are available in all of Atlantium's RZ163 system models.

SYSTEM START-UP

1. Check that the electrical cabinet's power cable is connected to the main power source.

2. Check that the Emergency Stop button (see Figure 2) is not pressed in. if it is, turn it clockwise to release it.

3. Lift all of the automatic circuit breaker switches to the upper (**On**) position (see Figure 1); the system will power up and the touch-screen monitor will come to life displaying the Atlantium software's opening screen (see Figure 2).

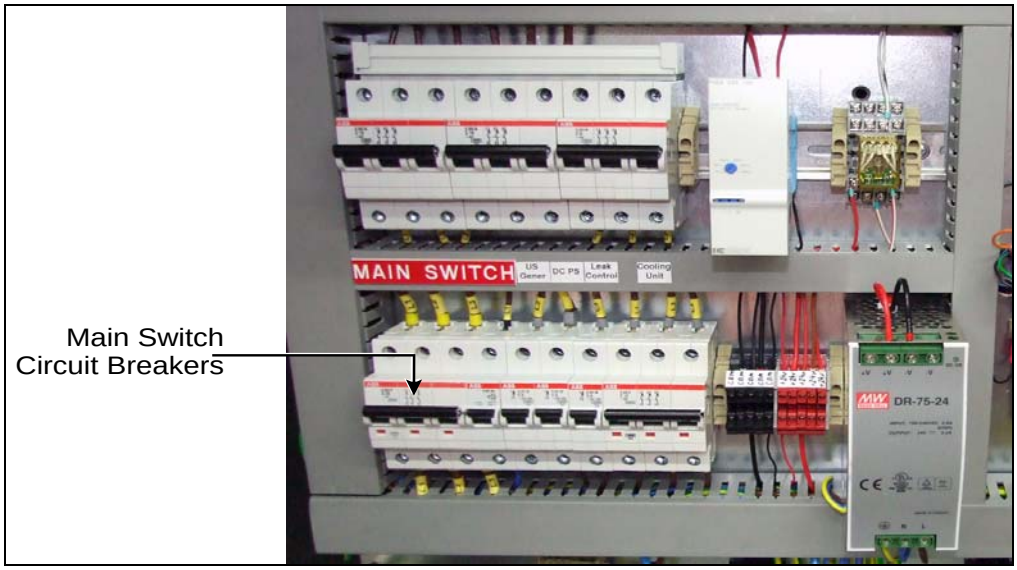


Figure 1: Automatic Circuit Breakers

Figure 2: Touch-Screen Monitor & Emergency Stop Button

ATLANTIUM SOFTWARE OPERATING SYSTEM

When the system is started, the Atlantium software's opening screen (see Figure 3) appears on the touch-screen display for a short time while the system performs its internal self-testing routines. During this time a progress bar will be running up on the bottom of the screen.

Figure 3: Startup Opening Screen

When the self-testing routines are completed satisfactorily, the **Main Operation** screen will appear (see Figure 4), displaying real-time information of the reactor's operating performance, including current operational status of each UV lamp in the reactor(s).

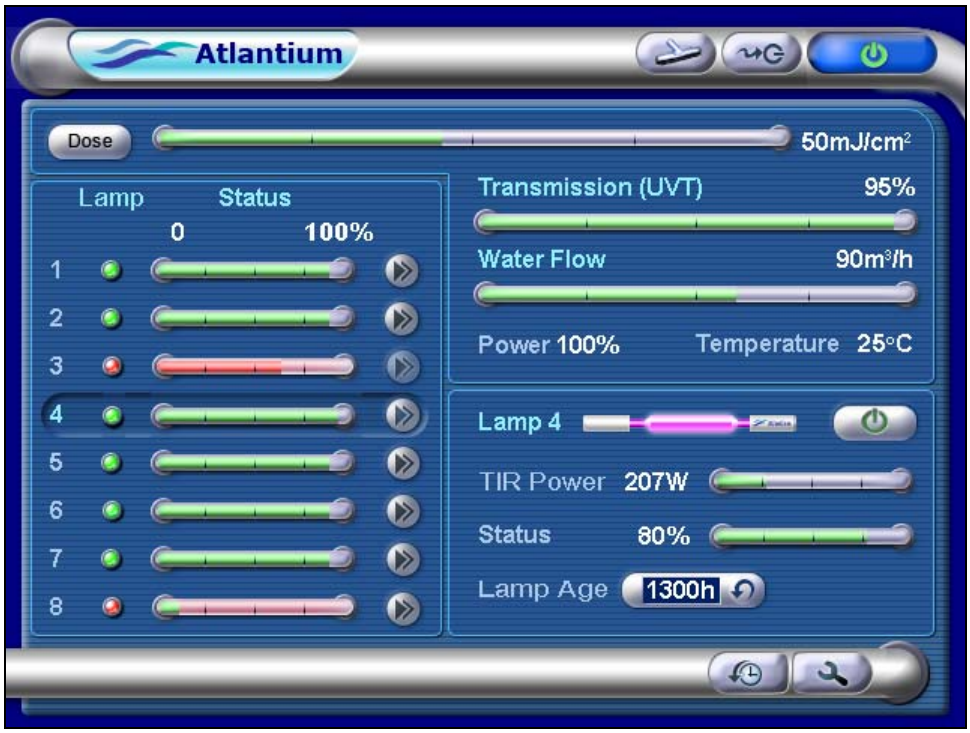


Figure 4: Main Operation Screen

Main Operation Screen

The **Main Operation** screen (see Figure 5) offers constant and real-time data concerning the operation of the reactor and its array of UV lamps. The screen also incorporates a few basic operation tools.

The elements of the screen are as follows:

1. **Main Reactor Operation Button** – touch this button to operate the Atlantium reactor(s): UV lamp(s), computer control of the system (UV power, etc.). Touch the same button again to turn the system off [1].

2. **Remote Operation Button** – when enabled (blue/green button), this feature allows the system to be turned on or off from the remote on/off switching mechanism. When the feature is disabled (gray/black button), the system may be turned on or off only from this screen.

> ***Important Note***
>
> This button appears on the main operating screen only if the **Remote On/Off** feature is activated in the **Settings** screen.

Figure 5: Main Operation Screen – Details

[1] Refer to page 9 for pop-up screen information.

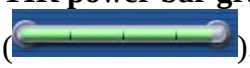
3. **Ultrasound Cleaning System Button** – this button operates the optionally-installed ultrasonic mechanism for cleaning the interior of the reactor's quartz tube.

> ***Important Note***
>
> This button appears on the main operating screen only if the optional ultrasound cleaning system is installed.

4. **Dose** – the currently applied UV dose is displayed in bar graph and text representations. The minimum and maximum allowable dose levels are user-defined (see **Operation Mode** on page 22), and are accessible for configuration by touching the **Dose** button. If the applied dose falls below or rises above the defined levels, the color of the bar graph changes from green to red.

5. **All Lamps At a Glance** – this portion of the screen displays the current status and UV TIR level of all the lamps in the system:

   - **Lamp status indicator** – green: satisfactory / red: less than optimal performance.

   - **TIR power bar graph** – shows the TIR level that the lamp is currently operating at ().
   - **Detailed lamp information** – touch this button for any of the available lamps to display that lamp's detailed information (see arrow # 10 in Figure 5 and refer to step 10 below).

6. **Transmission (UVT)** – this bar graph and numeric value exhibit the current quality of the water flowing through the system, in relation to the UV light traveling through the water. If the UVT percentage factor is high, then the clarity factor of the water is high, and vice versa. The lowest UVT percentage point where the system will shut down is user-defined (see **UVT Alarms** on page 15).

7. **Water Flow** – this bar graph and numeric value exhibit the current rate of water flowing through the reactor(s), measured in cubic meters per hour (m^3/h).

8. **Temperature** – of the water flowing through the reactor. If there are two branches, the displayed value is the calculated average of water temperature both reactors.

9. **Power** – displays a percentage value of the system's power.

10. **Detailed Lamp Information** – this portion of the screen offers the following UV lamp information and tools:

   - **On/Off button** – touch this button to toggle an individual lamp on or off [2].

   - **UV lamp on/off indicator** – = **On** / = **Off**.

[2] Refer to page 9 for pop-up screen information.

- **TIR power –** shows the TIR level that the lamp is currently operating at in bar graph and text representations ( ).
- **Status –** shows the operational status of the lamp in bar graph and text representations ( ).
- **Lamp Age –** displays the total number of hours the individual UV lamp has been operating for. Touch the button to reset the usage clock to zero after that particular lamp has been replaced with a new one.

11. **Settings Screen Access** – touch this button to go to the **Settings** screen (see page 10).

12. **Events Screen Access** - touch this button to go to the **Events** screen (see page 24).

Pop-Up Confirmation Messages

Several user-initiated actions will require confirmation before being completed. A pop-up window will appear with the message: **Are you sure?** Touch the **OK** button to complete the action (see Figure 6).

The actions that require confirmation are:

- **System Shutdown**
- **Lamp Shutdown**
- **Lamp Usage Clock Reset**

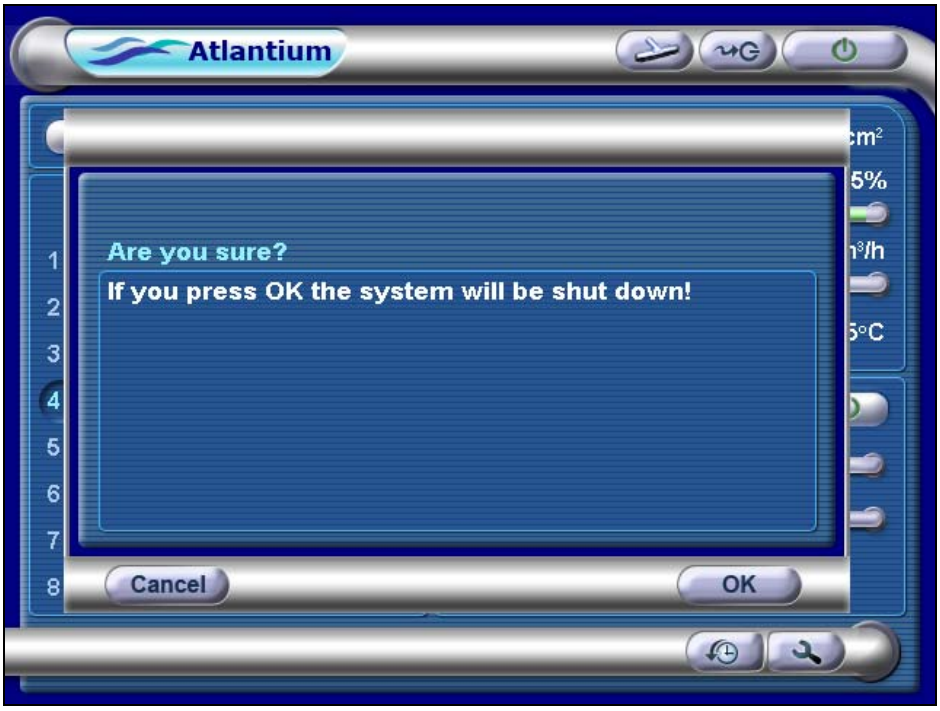


Figure 6: Pop-Up Confirmation Message (sample)

Settings Screens

To access the Settings screen touch the **Settings** screen icon on the **Main Operation** screen (see Figure 5); a **Password** pop-up screen will appear (see Figure 7).

Figure 7: Password Pop-Up Screen

Touch the numbered keys on the virtual keypad to enter the 4-digit password, and then touch the **OK** button; the **Settings** screen will appear on the touch-screen display (see Figure 8).

If you made an error entering the password, touch the backspace ( ) key to erase and re-enter the correct digits.

Important Note

The initial User and Service passwords will be delivered to the system's administrator by the Atlantium account executive or the service representative.

The **Settings** screen is exhibited in Figure 8. The serial numbers next to the various tags correspond to the settings pop-up windows appearing on the following pages.

Figure 8: Settings Screen

<u>Icon conventions for Settings screen and pop-up windows:</u>

- Setting is **Active**; touch once to deactivate

- Setting is **Inactive**; touch once to activate

- Opens a pop-up window with configurable settings

- **Normally Open** – one of the configurable settings in the pop-up windows; relates to the setting's electronic circuit. If the setting requires an open circuit to operate, this should be set to **Normally Open**.
- **Normally Closed** – one of the configurable settings in the pop-up windows; relates to the setting's electronic circuit. If the setting requires a closed circuit to operate, this should be set to **Normally Closed**.

- The dark blue side of the selector button is **Active**; the gray side is **Inactive**

- ▶ raises the % value; ◀ lowers the % value

Error Message and Actions:

There are four possible actions that may be configured, in some circumstances, as to how the system will react to an error message and these can be set in the pop-up windows:

- **None** – no error situation is present and no corrective action is required (no reaction).
- **Caution** – A short error message at the bottom of the **Main Operation** screen (see Figure 8).
- **Warning** – A pop-up window message, together with a remote alarm (if one is connected).
- **Shut-Down** – A pop-up window message with a remote alarm (if one is connected) **and** automatic system shut-down.

The following sections explain the various configurable settings in the Settings screen's pop-up windows

Remote On/Off (1)

The Atlantium system offers the option of setting up a remote on/off switching mechanism that will synchronize the reactor system to the facility. Example: the facility's water pump can be set up to send a signal to the reactor when water flow is started or stopped. When this setting is active the Atlantium system will recognize this signal and turn the reactor on or off, whichever is appropriate, without anyone having to manually turn the system on or off.

The pop-up window's configurable settings are (see Figure 9):

- **Normally Open**
- **Normally Closed**

Ensure that the **Active** button ( ) is pressed on the desired configuration and touch the **Close** button to return to the **Settings** screen (see Figure 8).

Figure 9: Remote On/Off Settings Pop-Up Window

Remote Alarm (2)

A remote alarm mechanism may be connected to the Atlantium system. When this setting is active and the system experiences an error situation, an alarm may be delivered to a remote destination in the form of a light or a bell.

The pop-up window's configurable settings are (see Figure 10):

- **Normally Open**
- **Normally Closed**

Ensure that the **Active** button ( ) is pressed on the desired configuration and touch the **Close** button to return to the **Settings** screen (see Figure 8).

Figure 10: Remote Alarm Setting Pop-Up Window

UVT Alarms (3)

The Atlantium system will deliver error messages when the water's UV transmission factor drops below a preset value.

The pop-up window's gauges and configurable settings are (see Figure 11):

- **Min UVT** – minimally acceptable UV transmission value (configure with the ◀ / ▶ buttons), below which the system will issue an alarm. The user sets this to warn of UVT degradation.

- **Action** – action performed if the factor falls below the **Min UVT** (configure with the ◀ / ▶ buttons)

Touch the **Close** button to return to the **Settings** screen (see Figure 8).

Figure 11: UVT Alarms Settings Pop-Up Window

Flow Meter (4)

Activate this configuration screen to integrate flow meter information into the Atlantium system.

The **Flow Meter** pop-up window's gauges and configurable settings are (see Figure 12):

- **Current Flow** – this gauge displays the current rate of water flowing through the system in cubic meters per hour
- **Flow Factor** - Read the flow meter's datasheet, find the meter's maximum-allowed flow rate, and enter that value divided by 100; i.e., if the max flow rate is 100 m^3/h, enter the value **1** as the Flow Factor (configure with the ◀ / ▶ buttons)
- **Max Flow** - a maximal water flow rate selector; if the actual flow rate exceeds this set value, the system will deliver an error message and alarm (configure with the ◀ / ▶ buttons)
- **Action** – action performed if the water flow rate exceeds the **Max Flow** (configure with the ◀ / ▶ buttons)

Touch the **Close** button to return to the **Settings** screen (see Figure 8).

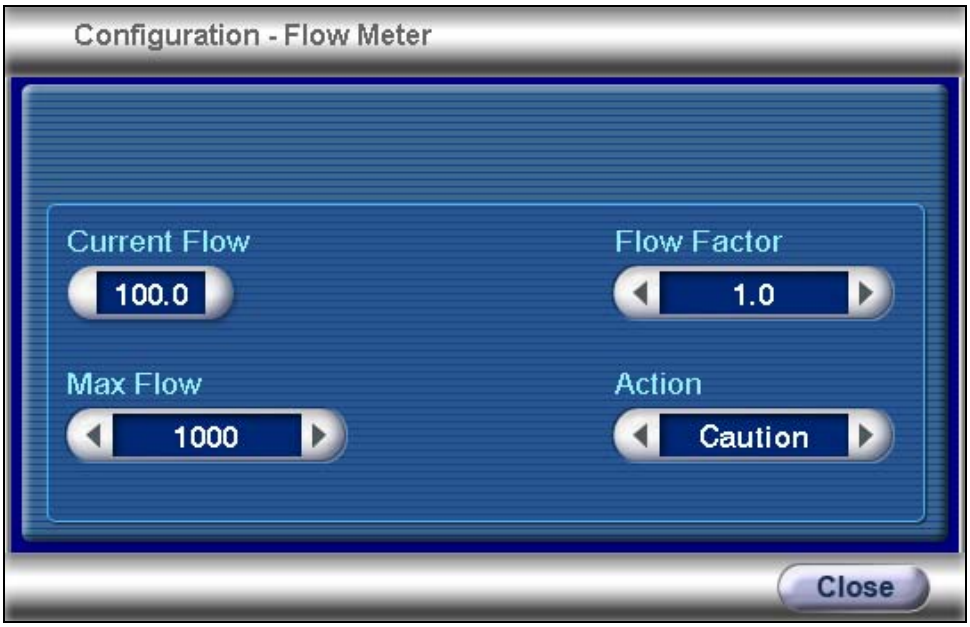


Figure 12: Flow Meter Configuration Pop-Up Window

Flow Switch (4)

Activate this configuration when a flow switch is connected to the system instead of a flow meter.

The **Flow Meter** pop-up window's configurable settings are (see Figure 13):

- **Normally Open**
- **Normally Closed**

Ensure that the **Active** button ( ) is pressed on the desired configuration and touch the **Close** button to return to the **Settings** screen (see Figure 8).

Figure 13: Flow Switch Configuration Pop-Up Window

Auto. Valves (5)

This is the area for setting the type of valves on the pipes that feed the reactor, and their level of automation. The facility may be equipped with electronically-controlled valves that will shut down the water supply to the reactor upon a signal from the Atlantium system (i.e. error situation), or may re-route the water supply to a different destination (i.e. water bypass system). The pop-up window's configurable settings are (see Figure 14):

- **Inlet Valve:** present or not present on the reactor
  - Inlet valve (if present): normally open (**NO**) or normally closed (**NC**)
- **Outlet Valve:** present or not present on the reactor
  - Outlet valve (if present): normally open (**NO**) or normally closed (**NC**)

If the facility has a water bypass system:

1. The **Bypass** button should be active ( ).

2. The pop-up window has two inputs for monitoring the polarity of the bypass valve:

- If the electronic polarity of the **bypass valve** is normally open, the **NO** radio button should be active ( ); if it is normally closed, the **NC** radio button should be active ( )
- If the electronic polarity of the **bypass valve's monitor** is normally open, the **NO** radio button should be active ( ); if it is normally closed, the **NC** radio button should be active ( ).

Ensure that the **Active** button ( ) is pressed on the desired configurations and touch the **Close** button to return to the **Settings** screen (see Figure 8).

Figure 14: Automatic Valves Configuration Pop-Up Window

Pressure (6)

Use this screen to correlate the specific pressure transmitter connected to the system.

Warning and alarms are hard-coded to alert the operator if the pressure in the reactor rises to within the danger range (8-10 bars).

The pop-up window's gauges and configurable settings are (see Figure 15):

- **Pressure** – this gauge displays the current water pressure inside the reactor
- **Pressure Factor** - read the pressure transmitter's datasheet, find the maximum-allowed pressure rating, and enter that value divided by 10; i.e., if the max pressure rating of the transmitter is 50 bars, enter the value **5** as the Pressure Factor (configure with the ◀ / ▶ buttons).

Touch the **Close** button to return to the **Settings** screen (see Figure 8).

Figure 15: Pressure Configuration Pop-Up Window

System Cleaning (7)

The ultrasound cleaning system[3] cleans the internal walls of the reactor with ultrasonic waves. If the Atlantium system is equipped with this system, this setting should be set to **Active** ( ).

The pop-up window's configurable settings are (see Figure 16):

- **Operate Every…:** operate the system at (user-defined) hourly intervals
- **Operation Duration:** operate the system for a (user-defined) number of minutes
- Note that the system can also be operated manually by touching the operation button at the top of the **Main Operating** screen (see Figure 5) or the **Settings** screen (see Figure 8) and then touching it again to stop the cleaning operation. This will not effect the automatic cleaning cycles established here.

Touch the **Close** button to return to the **Settings** screen (see Figure 8).

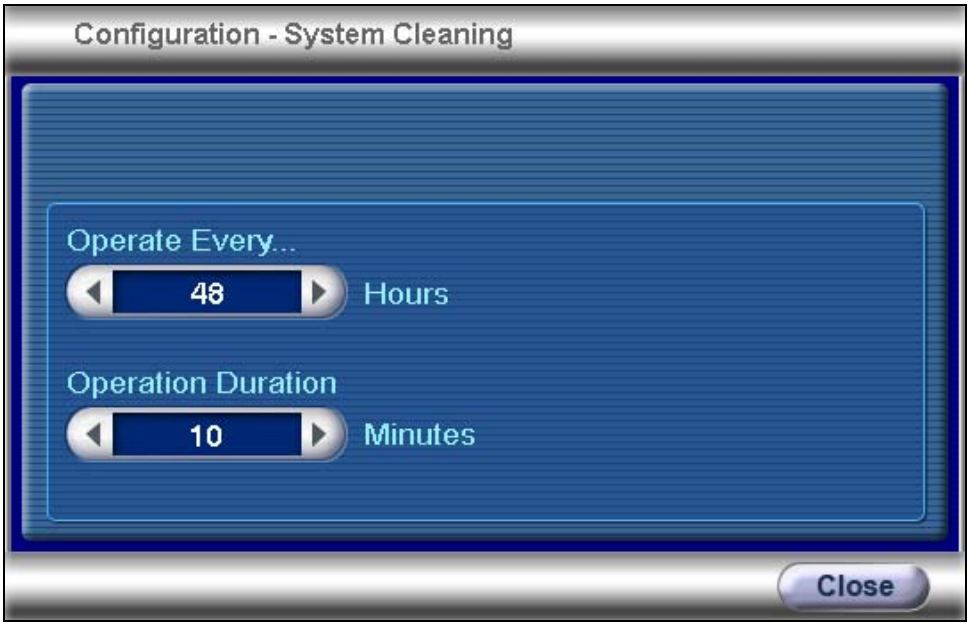


Figure 16: System Cleaning Configuration Pop-Up Window

Important Note

This software feature will be available only if the optional ultrasound cleaning system is installed.

[3] Optional Purchase Equipment

Lamps (8)

This pop-up window offers the possibility, in a multiple-lamp reactor, to shut down one of the UV lamps for station maintenance while leaving the other lamps in operation.

The lamps pop-up window's configurable settings are (see Figure 17):

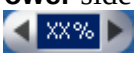
1. An operating lamp is represented by the **Enabled** ( ) icon.
2. A disabled lamp is represented by the **Disabled** ( ) icon.
3. After replacing a UV lamp, touch the button in the Lamp Age section of the **Main Operating** screen (see Figure 5) to reset the usage clock to zero.
4. **Auto Power On** – this feature restarts the system automatically when the main power supplied is renewed after a facility-wide power failure. Activate the feature ( ) for automatic restart, or deactivate ( ) for manual restart.

Ensure that the **Enabled** buttons are pressed on the desired lamps and touch the **OK** button to return to the **Main Operating** screen (see Figure 5).

Figure 17: Lamps Configuration Pop-Up Window

Operation Mode (9)

The Atlantium system generally has two alternate operational modes of control, but only the second is available for Validated Drinking Water for routine use (see Figure 18).

- **Power** – (see **A** in Figure 18) controls the level of power delivered by the UV lamps: touch-to-activate the **Power** side of the **Operation Mode** selector button and touch the ◀ or ▶ buttons of the  selector to set the desired power level; the system will now operate at a constant power level. This setting is used for CIP cleaning of the reactor (refer to the **Maintenance** guide).
- **Dose –** (see **B** in Figure 18) touch-to-activate the **Dose** side of the **Operation Mode** selector button and touch the **Dose Settings** button; the **Dose** configuration pop-up screen will appear.

The **Configuration - Dose** pop-up window's configurable settings are:

- **Minimum Dose mJ/cm^3** – configure with the ◀ / ▶ buttons
- **Dose Setpoint mJ/cm^3** – configure with the ◀ / ▶ buttons
- **Maximum Dose mJ/cm^3** – set the maximum UV dose (configure with the ◀ / ▶ buttons)
- **Action** (for each parameter) – action performed if the appropriate dose exceeds or falls below the set values (configure with the ◀ / ▶ buttons)

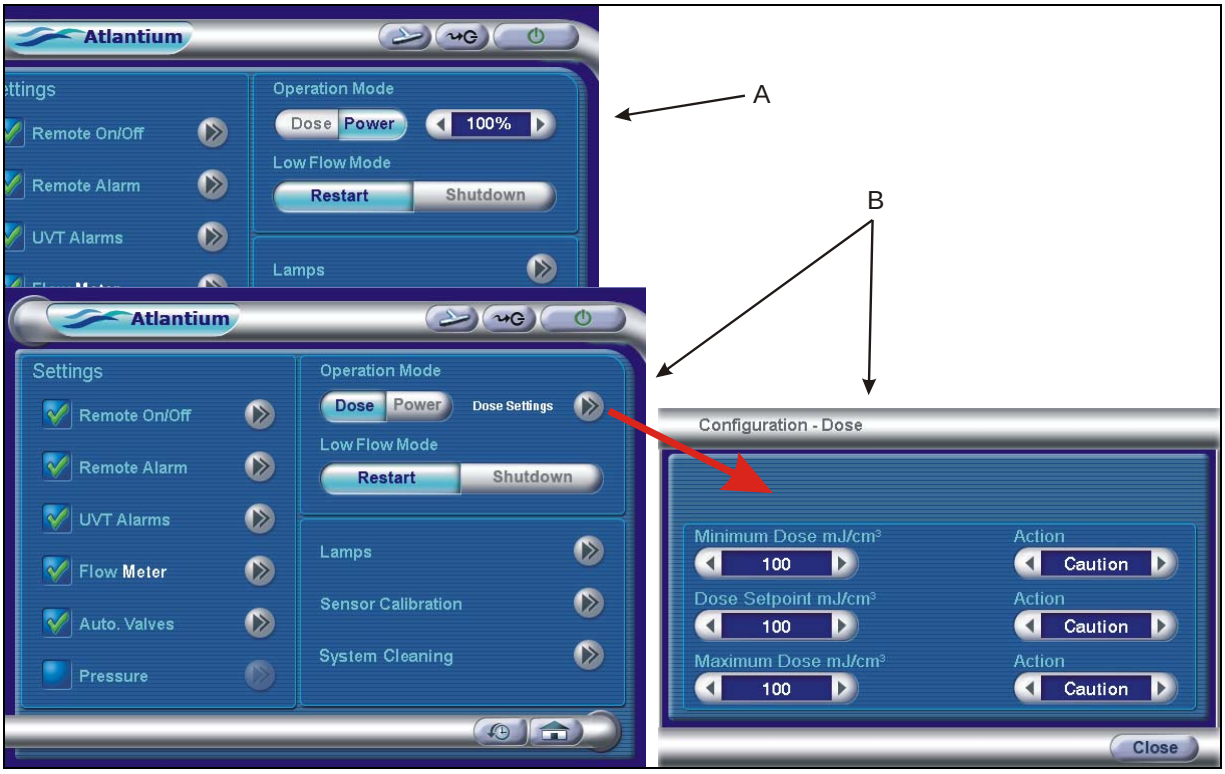


Figure 18: Dose & Power Settings and Dose Configuration Pop-Up Window

Low Flow Mode (10)

This selector sets the system's response if a low water flow situation occurs.

- **Restart** – if the water flow rate drops below the minimum flow rate set in the **Technician Settings** screen, for a time above the maximum flow delay set in the same screen, the **Low Water Flow** warning will be triggered, the UV lamps will automatically shut down and an event warning will be written to the log file. When the water flow rate returns to normal, the reactor will instantly return to full operation without the potential for delay that is typical with a conventional reactor.
- The default values for the minimum flow rate and maximum flow delay are set at the factory. The Atlantium technician may change these values during system installation based on local site requirements and conditions.

  If the low water flow condition persists for 45 minutes, the system will automatically shut down.
- **Shutdown** – The system views low flow as a sign of an upstream malfunction and shuts down in an orderly fashion. An alarm will be issued.

Error Message Area (11)

This area at the bottom of the **Main Operation** screen (see Figure 5) and the **Settings** screen (see Figure 8) is where error messages will appear, should an error situation occur.

Events Screen (12)

The **Events** screen (see Figure 19) is where you may view the data log of reactor events. Every defined event is written to the event log. The elements of the **Events** screen are as follows.

1. There are four Status levels of defined events, represented by the following icons:
2. **Informative Event** – this is a non-error type event, such as starting or stopping the system.
3. **Non-Fatal Event** – this is a non-fatal type error that does not automatically shut down the system.
4. **Non-Fatal Event Resolution** – if the system automatically resolves this error, the event icon will show a green ✓.
5. **Fatal Event** – this is a fatal error that shuts down the system; the error must be resolved by the operator, and the system restarted.
6. **Start Time** – date and time stamp of the beginning of the event.
7. **End Time** – date and time stamp of the end of the event.
8. **Details** – a short remark describing the event.
9. **Information Pop-Up** – touch this button to open a pop-up window with a detailed explanation of the nature of the event.

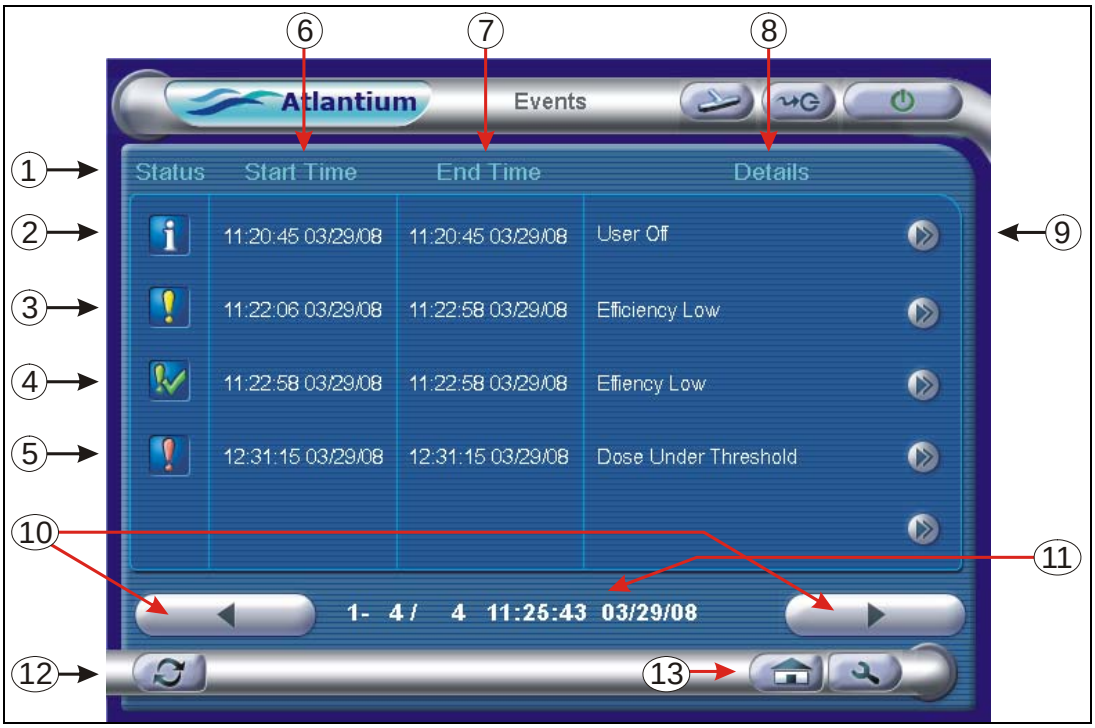


Figure 19: Events Screen

10. **Scrolling** – touch the ▶ or ◀ buttons to scroll up or down the list of events.

11. **Events Displayed & Current Time/Date Stamp** – the sample screen in Figure 19 is showing events # 1 through 4 / out of a total of 4 written events.

12. **Refresh** – touch this button to refresh the list of events, exhibiting the last five to be written to the log.

13. **Home** – touch this button to return to the **Main Operation** screen (see Figure 5).

Important Note

The **Event Log** database in the local controller is limited to 100 events. When the database is full, the latest event written will always cause the oldest one to be deleted.